

SELVAKUMAR MANOHARAN

AI/ML ENGINEER — DATA SCIENTIST

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PROFESSIONAL SUMMARY

AI/ML Engineer (Machine Learning Engineer) and Data Scientist with **2.5 years of hands-on experience** building machine learning and computer vision solutions across digital identity verification, KYC/fraud risk, and document compliance for banks — including face verification, face recognition, liveness detection, OCR-based document verification, and behavioral analytics. Proficient in Python, SQL, and the full model lifecycle — data preprocessing, training, evaluation, threshold calibration, and deployment via ONNX and FastAPI. Also skilled in Generative AI, including RAG, LangChain, LangGraph, Hugging Face, and prompt engineering. Postgraduate Diploma in Data Science and Analytics, University of Hertfordshire.

TECHNICAL SKILLS

Programming: Python, SQL, Pandas, NumPy, Data Cleaning, Data Preprocessing, Feature Engineering, Data Pipelines, Web Crawling (Playwright, Selenium, Requests)

ML & Deep Learning: Machine Learning, Supervised Learning, Classification, Regression, Model Evaluation, Model Validation, Hyperparameter Tuning, Model Calibration, Neural Networks, PyTorch, Scikit-learn, OpenCV, Computer Vision, Image Preprocessing, Image Classification, Transfer Learning, Embeddings, Similarity Matching, OCR, Document Data Extraction

Generative AI: LLMs, Prompt Engineering, LLM-Assisted Data Extraction, RAG, LangChain, LangGraph, Hugging Face, NLP

Data Analytics: Data Analytics, Power BI, Microsoft Excel, Data Visualization, Statistical Analysis, Dashboards, Exploratory Data Analysis, Google Analytics, Google Tag Manager

APIs & Deployment: FastAPI, REST APIs, API Integration, Asynchronous Processing, JSON Schema Design, ONNX Runtime, Model Inference, Model Deployment, Docker, AWS, Git, GitHub

Dev Tools: Cursor, Claude Code, Jira

PROFESSIONAL EXPERIENCE

Xylum Global Services JUL 2025 — PRESENT | Chennai, India

Data Analyst / Data Scientist

- Engineered production computer vision pipelines powering **real-time digital identity verification**, covering face detection, alignment, landmark extraction and pose estimation with ONNX-served embedding models, reducing per-request inference time by **~35%** for **onboarding/KYC use cases**.
- Trained and validated deep learning classifiers for face liveness and presentation attack detection, delivering a deployable **anti-fraud** model distinguishing genuine faces from spoofs across multiple attack types.
- Calibrated decision thresholds for verification and liveness models, balancing FAR/FRR trade-offs to meet **production risk and compliance tolerance** — a direct input to **fraud-risk and regulatory decisions**, not just model accuracy.
- Built OCR-based extraction for **Tamil Nadu** government land record documents (Patta, EC, FMB) — a **document compliance check for bank loans** — integrating government source APIs and cross-verifying fields, surfacing mismatches and risk scores that replaced a previously manual **title-verification process** and cut document review time by **~40%**.
- Designed an API-first behavioral fingerprinting service for **fraud and trust-risk scoring**, converting raw user interaction signals into structured session-level features for downstream risk profiling.
- Integrated models into production through FastAPI REST services and Docker packaging, maintaining version-controlled code in Git/GitHub with Cursor and Claude Code to accelerate development.

CloudHalo Technology Services APR 2024 — JUN 2025 | Chennai, India

Software Engineer / Data Analyst

- Built LLM-assisted web crawling pipelines with Playwright and Selenium, applying prompt-driven LLM parsing to convert unstructured pages into schema-consistent structured data, eliminating manual gathering across multiple source sites.
- Automated recurring collection through scheduled headless browser sessions, cutting data refresh time by **~30%** and reducing maintenance when source layouts changed.
- Built EDA-driven visualizations and dashboards, enabling stakeholders to act on insights without ad hoc queries.

CloudHalo Technology Services NOV 2023 — APR 2024 | Chennai, India

Software Engineer Trainee (Intern)

- Built Python data extraction scripts using Requests and BeautifulSoup to collect structured data from web sources, replacing manual collection with repeatable pipelines and cutting collection time by **~30%**.
- Cleaned and preprocessed raw datasets with Pandas and NumPy, producing analysis-ready tables that supported downstream reporting.

PROJECTS

Face Recognition and Liveness DEEP LEARNING · COMPUTER VISION · ONNX

- Verification & Recognition** — Engineered end-to-end verification (detection, alignment, landmarks, embedding generation and similarity matching), delivering a single API call returning a match decision with confidence score. Benchmarked embedding models, migrated backbone to AuraFace, consolidated inference code and calibrated similarity thresholds before integrating ONNX inference behind a REST API.
- Liveness & Anti-Spoofing** — Prepared the presentation attack dataset with standardized capture conditions across genuine and spoof samples. Trained classifiers, analyzed misclassifications and calibrated operating threshold, producing a production-ready anti-spoofing gate deployed alongside the verification service.

Unified Patta-EC Document Verification API OCR · API INTEGRATION · RISK SCORING

- Built the OCR extraction layer for government Patta, EC and FMB documents, recovering structured Tamil-language fields and converting unstructured government PDFs into queryable, validated data.
- Integrated government source APIs with field-level similarity scoring and match thresholds, surfacing mismatches that replaced manual cross-checking. Authored versioned REST API spec covering async endpoints and error handling.

Behavioral Fingerprinting Platform DATA SCIENCE · FEATURE ENGINEERING · REST API

- Designed an API-first solution transforming interaction signals into structured behavioral features, with event schemas, session aggregation and derived attributes exposed through REST endpoints for multiple consuming services.

EDUCATION

University of Hertfordshire, Hatfield, United Kingdom FEB 2021 — JAN 2023

Postgraduate Diploma, Data Science and Analytics

SRM TRP Engineering College, Trichy, India JUL 2015 — MAY 2018

B.E. Electronics and Communication Engineering

AWARDS & CERTIFICATIONS

Certificate of Appreciation — Outstanding Performance Xylum Global Services · AUG 2025 — NOV 2025

The Complete Python Bootcamp: From Zero to Hero in Python Udemy, Pierian Training · Oct 2024 · [Verify certificate](#)